▸ VALENTÍN ANDRADA, JOSÉ LUIS CASTIGLIONI, WILLIAM ZULUAGA BOTTERO, *Translations and internal algebras over a category.*
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Following Moraschini [2] and earlier ideas of McKenzie [3], 'general' classes of algebras X, Y were introduced together with a notion of 'translation' between semantic consequence relations $\models_Y$ and $\models_X$, subsuming classical translations such as Kolmogorov's and Glivenko's. It is shown there that such general translations correspond to pairs of adjoint functors between X and Y. This provides a concrete description of the adjunction and presents the associated translation explicitly through several examples of classical functorial constructions, such as those of the 'Kalman-type' and 'regular-elements-type', thereby establishing a framework connecting two distinct approaches to the same phenomenon.

Inspired by Bergman [1] and following Moraschini [2], we start from a general class of algebras X and investigate a correspondence between internal X-algebras over a suitable category $\mathcal{C}$ and right adjoint functors $\mathcal{C}^{\mathrm{op}} \to X$. In this talk we present the general ideas underlying this approach and describe partial results toward establishing such a correspondence.

[1] George M. Bergman, *An invitation to general algebra and universal constructions*, ***Universitext***, Springer, New York, 2015.

[2] Tommaso Moraschini, *A logical and algebraic characterization of adjunctions between generalized quasi-varieties*, ***Studia Logica***, vol. 109 (2021), no. 5, pp. 1045–1080.

[3] Ralph McKenzie, *An algebraic version of categorical equivalence for varieties and more general algebraic categories*, in ***Logic and Algebra*** (Pontignano, 1994), Lecture Notes in Pure and Applied Mathematics, vol. 180, pp. 211–243, Marcel Dekker, New York, 1996.